

Novel Feed-Forward Technique for Digital Bang-Bang PLL to Achieve Fast Lock and Low Phase Noise

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Abstract—This paper presents a novel technique to reduce the locking time in Digital Phase-Locked Loop (DPLL) based on Bang-Bang Phase Detector (BB-PD). The implemented 65-nm CMOS fractional-N frequency synthesizer generates an output signal between 3.7 and 4.1 GHz from a 52 MHz reference clock and improves the trade-off between phase noise, due to the loop quantization, and locking time, exploiting a digital locking loop that avoids look-up table (LUT) and finite state machine-based (FSM) locking schemes. Measurements show that the output signal spot noise at 20 MHz from the carrier is -150.7 dBc/Hz while the best locking time, for a coarse step of 364 MHz, is 115 μ s, overcoming the locking time limitations and avoiding cycle slips that usually affect the 1-bit phase detector PLL.

Index Terms—Digital phase-locked loop (DPLL), digital to time converter (DTC), frequency synthesis, locking aid, locking time, phase noise.

I. INTRODUCTION

FREQUENCY synthesizers requirements for new radio communications standards are becoming more and more demanding especially in terms of area, phase noise and power consumption. Thanks to their versatility and scalability the Digital Phase-Locked Loop (DPLL) architecture has reduced the loop filter area occupation in frequency synthesizers, providing at the same time a digital correction to the analog impairments using background digital calibration algorithms. A basic DPLL scheme is depicted in Fig.1 where the standard frequency synthesizer blocks are assisted by the intensive use of digital algorithms, implemented with the standard-cell

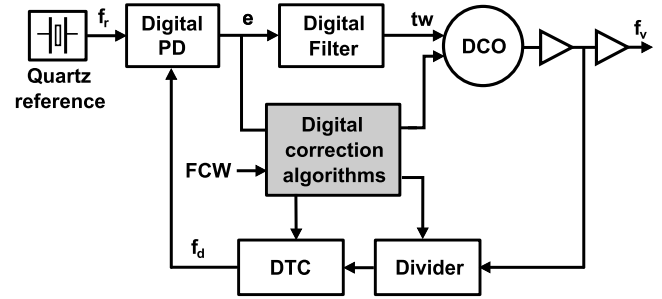


Fig. 1. Simplified scheme of a digitally intensive fractional-N frequency synthesizer based on a Bang-Bang phase detector (BB-PD).

methodology. This approach is made possible thanks to the availability of the phase error e , produced by a time-to-digital converter (TDC), in the digital domain.

Among the DPLL architectures, the Bang-Bang DPLL (BB-DPLL) ones have further reduced the power consumption and the power of the fractional spurs [1]. In this architecture, the phase detector linearity requirement, that is usually needed to guarantee low-spur power level, is moved to the digital-to-time converter (DTC) in the feedback path and can be easily achieved by digital pre-distortion LMS-based algorithms [1]. In this way the phase detector can be designed with just 1-bit, reducing area and power consumption with respect to the multi-bit TDC approach.

Beside the advantages given by the digital algorithms, the quantization introduces a trade-off between locking time and PLL output signal phase noise. As will be discussed in the next chapters, by increasing the period gain of the digitally controlled oscillator (DCO) the locking transient speeds-up, but at the same time the quantization noise at steady state increases. This trade-off is further exacerbated by the limited phase detector dynamic range in a BB-DPLL. Due this intrinsic limitation a BB-DPLL presents very long locking transients, around tens of milliseconds, unacceptable for most of the commercial applications.

Many approaches have been proposed in literature to reduce the settling time. A common solution is to increase the loop bandwidth during the locking transient [2]–[4]. This technique is usually applied in analog PLLs where it is implemented by

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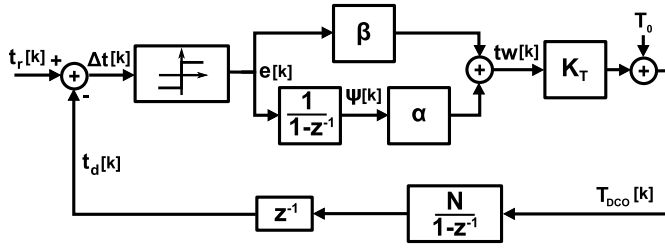


Fig. 2. Simplified discrete-time single-time-domain linear model of a BB-DPLL.

increasing the charge-pump current or reconfiguring the loop filter. But changing the loop filter dynamically could lead to loop instability and additional phase error. Another method is to use a Frequency Locked Loop (FLL) based on a background counter block [7]. However, the solution is power consuming, especially with high frequency output signals. Knowing the DCO tuning characteristic it is also possible to store the tuning settings in a LUT [5], [6], but this method does not take account of process-voltage-temperature variations (PVT), requiring a periodic characterization of the DCO tuning range. A more flexible solution is based on a finite state machine (FSM) [8]–[12], where a searching algorithm find the DCO control word to minimize the frequency and the phase errors with respect to the desired frequency. However, FSM searching algorithms are not always fail-safe and require some time to achieve a proper lock over a wide tuning range.

In this paper we discuss a new feed-forward locking technique for BB-DPLL that overcomes the dynamic range limitation of the BB-PD, reducing the locking time without altering the quantization noise and therefore the overall noise performance of the system. The results, already presented in [15], are extended by adding new quantitative guidelines, design consideration and measurements.

The paper is structured as follows. In Section II, starting from [14], the trade-off between locking transient and noise performance in a single loop BB-DPLL is derived. New considerations and results regarding the cycle slip phenomena are added. Section III shows a possible BB-DPLL architecture with fast locking, based on multiple parallel DCO control loops, to overcome the single bit phase detector locking time limitations. Theoretical results are compared against simulations, highlighting the practical implementation issues of this approach. Section IV describes the implemented feed-forward locking-aid scheme and the advantage of using this technique with respect to the multiple parallel DCO control loops. Finally, Section V presents the measured locking time performance of the BB-DPLL implementing the proposed technique, extending the results already discussed in [15].

II. SINGLE LOOP BBPLL

A. Re-Locking Time and Cycle Slips

The frequency synthesizer depicted in Fig. 1 can be simplified into a discrete-time single-time-domain model as in Fig. 2. The frequency control word (FCW) sets the divider modulus to N and the BB-PD is modelled by a non-linear ± 1 block.

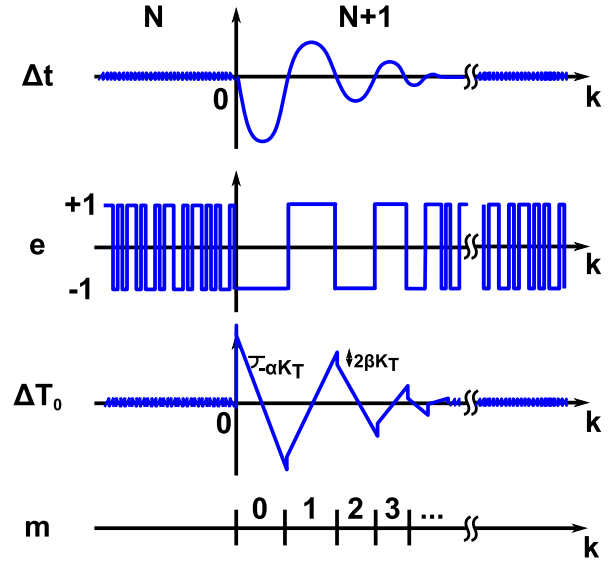


Fig. 3. Locking transient of main loop variables in a BB-DPLL.

The variables $t_r[k]$ and $t_d[k]$ denote the leading-edge timestamps of the reference and the divider signals, respectively. The constant T_0 is instead the DCO free-running period (i.e., the inverse of the free-running frequency f_0) [14]. Each variable is sampled at the reference clock rate and the index k counts the reference clock cycle number. At each timestamp, the state of the system is fully represented by two state variables: the time error $\Delta t[k] = t_r[k] - t_d[k]$ and the output of the integration branch, $\Psi[k]$, or, in alternative, by the period error, $\Delta T_0[k]$, which is the difference between the actual DCO period, $T_{DCO}[k]$, and the steady state value, $T_{DCO}[+\infty]$. It is

$$\Delta T_0[k] = T_{DCO}[k] - T_{DCO}[+\infty]. \quad (1)$$

Any time the divider modulus changes from N to $N + 1$, the initial DCO output period is longer than its final steady state value, $T_{DCO}[+\infty]$. As shown in Fig. 3, the period error $\Delta T_0[k]$ is positive, thus leading to a negative phase error $\Delta t[0] = t_r[0] - t_d[0] < 0$. The BB-PD digital output becomes $e[0] = -1$ and the state variable $\Psi[k]$ is reduced. The feedback makes the period error, $\Delta T_0[k]$, decreasing linearly at a pace αK_T , while $\Delta t[k]$ follows a parabolic trend. The detailed dependence is derived in Appendix A leading to (23) with a peak amplitude given by (35), that can be also written as

$$\Delta t_{max} = \frac{1}{R} \frac{N}{2} \left(\frac{\Delta T_0^2}{\beta K_T} \right), \quad (2)$$

where $R = \alpha/\beta$ is a parameter related to the loop stability and ΔT_0 is the DCO period error at the begin of the first trajectory section, $m = 0$. After peaking, $\Delta t[k]$ decreases until it becomes positive. At this point, $e[k]$ switches to $+1$ and $\Psi[k]$ is now increased following a new trajectory section $m = 1$. The transient goes on through different locking trajectory sections until the new steady state is reached. Starting from the trajectory equations presented in [14], in Appendix A it is

shown that the locking time can be approximated by

$$T_{lock} \approx T_{ref} \frac{1}{R(2-R)} \left(\frac{\Delta T_0}{\beta K_T} \right)^2, \quad (3)$$

where T_{ref} is the reference period. However, both R and (βK_T) are set by other considerations and limited room is left to the designer to minimize the locking time without affecting other performance.

The loop gain of the model system in Fig. 2 has been extensively discussed in [13]. The PLL bandwidth is approximately given by

$$f_{BW} \approx \frac{f_r}{2\pi} (N\beta K_T K_{BB}), \quad (4)$$

where $f_r = 1/T_{ref}$ and the average BB gain, K_{BB} , is given by

$$K_{BB} = \sqrt{\frac{2}{\pi}} \cdot \frac{1}{\sigma_{\Delta t}}, \quad (5)$$

where $\sigma_{\Delta t}$ is the rms value of the jitter at the BB-PD input. To guarantee enough phase margin, also considering PVT variations, the zero frequency of the loop,

$$f_z \approx \frac{f_r}{2\pi} R, \quad (6)$$

should be chosen to be one decade less than the PLL bandwidth f_{BW} [16]. The R value is accordingly set to be

$$R = \frac{\alpha}{\beta} \ll \frac{2\pi f_{BW}}{f_r 10}. \quad (7)$$

The βK_T value affects, instead, the steady state output jitter value. In fact, under the assumption that the random noise of the reference path is dominant, it is $\sigma_{\Delta t}^2 \approx \sigma_{tr}^2$. The system at steady state is in a random noise regime [19] and the minimum output jitter is reached for [16]

$$\beta K_T \lesssim \frac{\sqrt{3}}{N} \sigma_{tr}. \quad (8)$$

It follows that in the standard BB-DPLL architecture, an intrinsic trade-off exists between the output noise and the locking transient. To speed-up the locking transient, the gain βK_T should be increased but at the expense of a higher jitter. Such a trade-off is particularly critical for low phase noise applications. For example, let us consider a BB-DPLL with parameters similar to those adopted in the implemented stage. The reference frequency is 52 MHz with a phase noise of -146 dBc/Hz , corresponding to $\sigma_{tr} = 1.6 \text{ ps}$. The DCO period resolution is $K_T = 1.11 \text{ fs/LSB}$ with a phase noise of -126 dBc/Hz at 1 MHz from the carrier. With $N = 74$, the output steady state frequency is 3.848 GHz. Using the loop filter parameters $\alpha = 2^{-12}$ and $\beta = 2^{-1}$, therefore $R = 2^{-11}$, the estimated is about BB-DPLL bandwidth is around 150 kHz with an output jitter around 150 fSRMS.

With these settings a frequency step Δf_0 of just 100 kHz would have a locking time around 2.9 ms. This value is estimated based on (3) and using the approximation $\Delta T_0 = \Delta f_0 / f_0^2$. By increasing the frequency step Δf_0 , the locking time T_{lock} increases following (3), as depicted in Fig. 4.

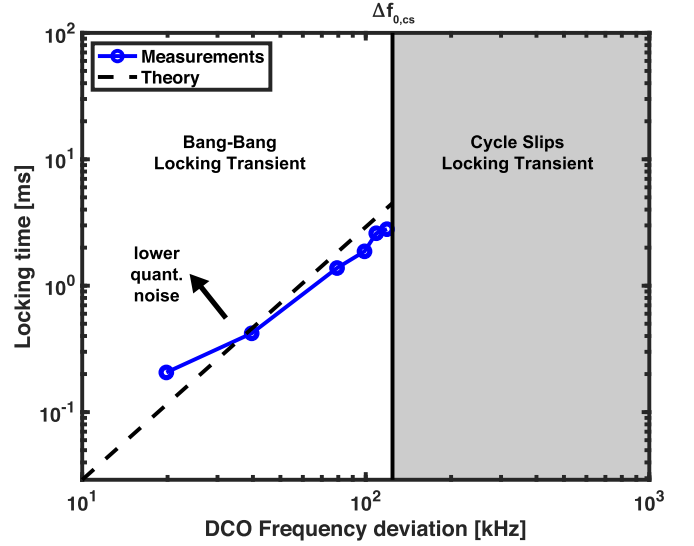


Fig. 4. Locking time of the BB-DPLL by changing the initial DCO period error: theory form (3) (black dashed line), and measurement results (blue line) on the implemented BB-DPLL.

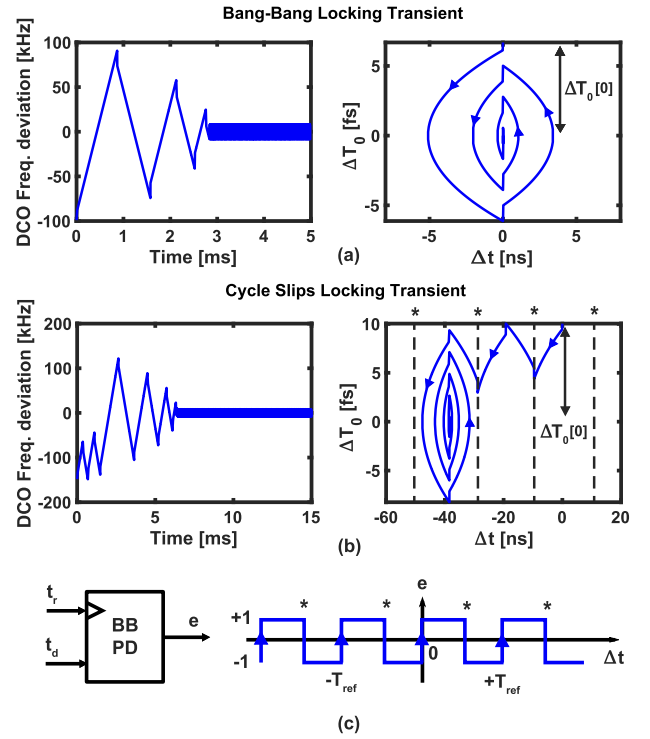


Fig. 5. Locking transient for an initial DCO period error $\Delta T_0[0]$: (a) lower than $\Delta T_{0,cs}$, (b) larger than $\Delta T_{0,cs}$, and (c) the BB-PD periodic characteristic. The dashed lines marked with (*) in (b) are the loop instability points that occur at each BB-PD characteristic falling edge in (c).

The trend, which is close to the values measured on the fabricated stage (blue line), stops around 100 kHz (grey area) where the BB-DPLL transient begins to be characterized by cycle slips. To explain this regime Fig. 5 (a) shows a simulated transient of the period error, $\Delta T_0[k]$, which follows the model transient already described in Fig. 3. The figure also shows the system trajectory over the $\Delta T_0[k]$, $\Delta t[k]$ state-space.

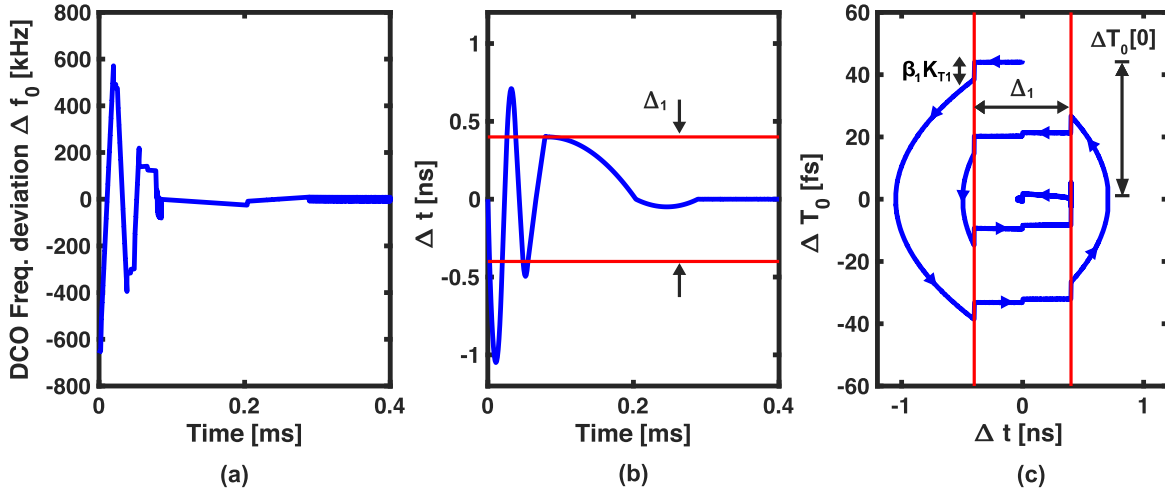


Fig. 6. Locking aid trajectory in a multi-loop architecture, with $\Delta f_0 = 600 \text{ kHz}$: (a) Output frequency locking transient; (b) Δt locking transient; (c) locking trajectory around the final steady state point.

Note the peculiar parabolic branches followed to recover the initial period error $\Delta T_0[0]$. The peak time error, Δt_{max} , of the locking parabolic trajectory, starting from an initial DCO period deviation $\Delta T_0[0]$, is given by (35). Based on this result, it turns out that by increasing $\Delta T_0[0]$ the peak time error Δt between the t_r and t_d signals may exceed $T_{ref}/2$. When this condition the BB-PD output changes from +1 to -1 and the sign of the feedback loop becomes positive. The effect is due to the periodic behaviour of the phase detector characteristic depicted in Fig. 5 (c).

The dashed vertical lines in Fig. 5 (b) are instability points. The figure also shows the trajectory for $\Delta f_0 = 150 \text{ kHz}$. The first parabolic trajectory exceeds the instability boundary. Both the DCO period $T_{DCO}[k]$ and $\Delta t[k]$ increase and locking is eventually reached again, but after some cycle slips at one of the following rising edges of the reference clock. The initial DCO period error leading to a cycle slip, $\Delta T_{0,cs}$, can be derived from (2) by taking $\Delta t_{max} = T_{ref}/2$. It is

$$\Delta T_{0,cs} = \sqrt{RT_0(\beta K_T)}. \quad (9)$$

The corresponding DCO frequency deviation is $\Delta f_{0,cs} = \Delta T_{0,cs}/T_0^2$. For the single-loop BB-DPLL with the parameters listed above, set to get an optimum output jitter, $\Delta f_{0,cs}$ is only about 124 kHz , which is unacceptable for practical applications.

In conclusion, the single loop BB-DPLL architecture does not have enough degrees of freedom to optimize both phase noise and locking transient duration. On one side, according to (3), βK_T must be increased to improve the locking time. On the other side, based on (8), the jitter degrades.

III. MULTI-LOOP ARCHITECTURE FOR FAST LOCKING

A. Locking Performance

One way to break this trade-off is to use two separate loops: one designed to guarantee minimum jitter in the steady state random noise condition and the other to speed-up the locking transient.

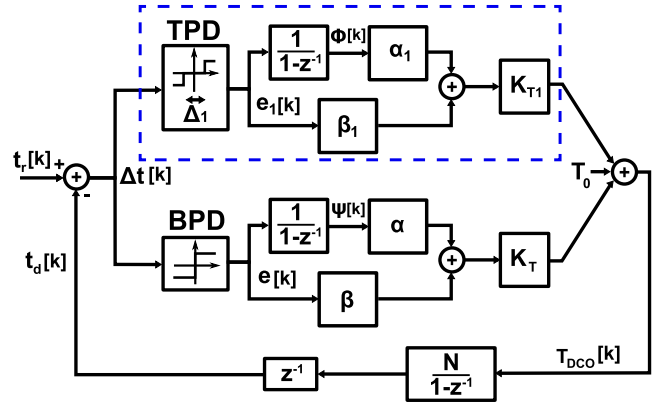


Fig. 7. Multi-loop BB-DPLL architecture model with the locking-aid path in evidence.

Fig. 7 shows the corresponding architecture where an additional ternary phase detector (TPD) is adopted. At steady state the phase error is inside the TPD dead-zone $-\Delta_1/2 \leq \Delta t \leq \Delta_1/2$ and the additional phase detector is idle. Outside the dead-zone the phase detector behaves as the BB-PD controlling with a digital PI filter a coarser DCO bank. This additional path, denoted as locking-aid branch, can be sized with a larger $\beta_1 K_{T1}$ value, thus allowing a fast transient without compromising the output phase noise at steady state.

Fig. 6 shows a locking transient in this architecture. The outer loop is triggered if the error of the DCO period at the beginning of the transient, ΔT_0 , is large enough to make the trajectory reaching a peak Δt time error larger than $\Delta_1/2$. This threshold can be evaluated using (2), and it is

$$\Delta T_{0,faid} = \sqrt{R(\beta K_T) \frac{\Delta_1}{N}}. \quad (10)$$

Using this result it is possible to split the locking time in two parts, one controlled by the outer loop and one controlled by the main loop. Appendix B reports the detailed calculations. The overall lock-time can be approximated by the sum

of two terms

$$t_{lock} \approx T_{ref} \frac{1}{R_1(2-R_1)} \frac{\Delta T_0^2 - \Delta T_{0,faid}^2}{(\beta_1 K_{T1})^2} + T_{ref} \frac{1}{R(2-R)} \left(\frac{\Delta T_{0,faid}}{\beta K_T} \right)^2. \quad (11)$$

The first term is related to the locking transient controlled by the outer loop. The second corresponds instead to the settling time within the inner dead-zone, Δ_1 , only driven by the inner loop.

The approximation consists in neglecting multiple crossings of the dead-zone, during the first part of the locking trajectory, but, as it will be seen in the following, the estimate given by (11) is reasonably accurate. It follows that, starting from the gain βK_T compatible with optimum jitter performance in (8), the first contribution to the locking time can be pushed in the sub-millisecond range by reducing the dead-zone amplitude, Δ_1 . The second term is instead minimized by increasing $\beta_1 K_{T1}$, which is now decoupled from the jitter performance.

The dead-zone amplitude, Δ_1 , cannot be reduced indefinitely. As suggested in [22], Δ_1 should be much larger than $6\sigma_{\Delta t}$ to avoid random triggering of the locking-aid loop and the optimum value for the locking transient is $\Delta_1 \approx 2\beta_1 K_{T1} N$. Moreover the K_{T1} gain should guarantee the tuning characteristic overlap and for this reason its value is related to the K_T gain.

Taking the values of the previous example, with $R = 2^{-11}$ and $\Delta_1 = 800$ ps the first contribution to the locking time is about 200 μs . Moreover, we may take $\beta_1 = 2^{-1}$ as in the main loop, $\alpha_1 = 2^{-7}$, for $R_1 = 2^{-6}$, and a period gain K_{T1} of the coarser DCO capacitive bank equal to 10.9 fs/LSB. With these values the second term contributes to the locking time for less than 200 μs also for a frequency jump of about 1 MHz. Moreover, thanks to the higher $\alpha_1 K_{T1}$ gain, the cycle slip threshold is moved to a higher frequency error. From (9), but using the $\alpha_1 K_{T1}$ value in place of αK_T , the new cycle slip threshold is shifted around 2.2 MHz, that is one decade higher with respect to the BB-DPLL without locking-aid techniques.

Fig. 8 shows the comparison between simulation results and (11) for the BB-DPLL with the parameters already listed above. Despite the approximations, (11) is able to fit the simulated multi-loop locking time behaviour. When the locking-aid loop is turned on, for a frequency error of 26 kHz ($\Delta_1 = 800$ ps), thanks to its higher gain, the total locking time does not increase too much. The discrepancy between the simulated values and (11) comes from the assumption of neglecting the repetitive crossing of the dead-zone.

IV. THE FEED-FORWARD BRANCH

The solution proposed in Section III considers an ideal DCO with no capacitive bank quantization and an infinite tuning range. In real implementations these assumptions fail. The finite DCO tuning range is segmented in more capacitive banks, composed by finite capacitive elements that are properly designed to optimize the overall quality factor. Moreover, each bank has a gain related to the other banks to keep an overlap between the tuning characteristics. The quantization

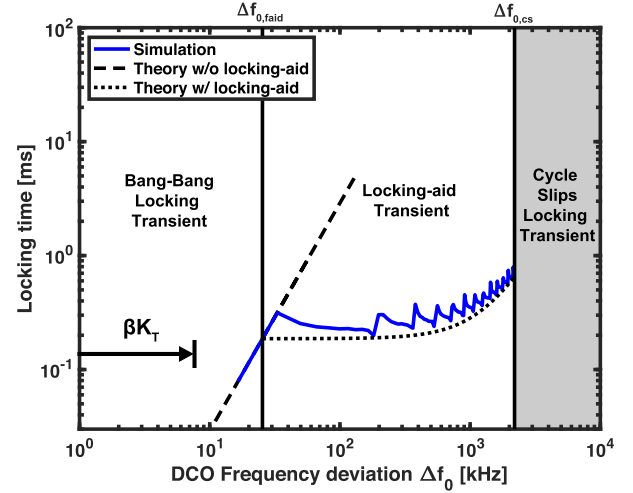


Fig. 8. Comparison between the locking time estimated for the BB-DPLL without multi-loop locking-aid (dashed line) and with multi-loop locking-aid architecture (dotted line). The blue line shows the simulation results.

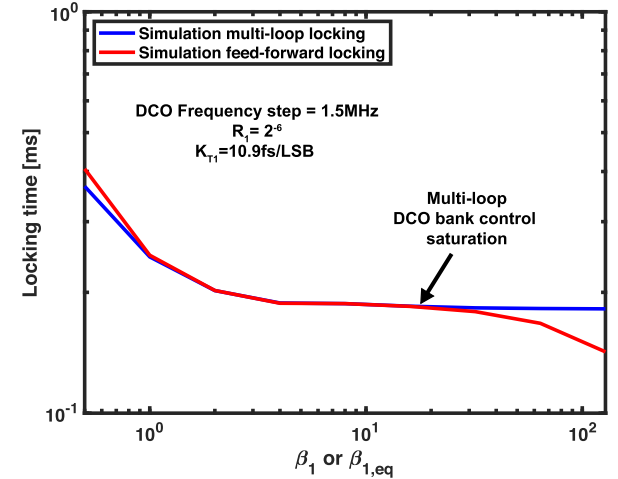


Fig. 9. Comparison of the locking time in a multi-loop architecture and in a feed-forward architecture for a $\Delta f_0 = 1.5$ MHz frequency step, varying β_1 or the $\beta_{1,eq}$ value while keeping R_1 constant.

of the capacitive bank, the limited number of capacitive cells and the limited K_{T1} gain, due to the need to maintain the overlap between the tuning characteristics, are detrimental for the locking performance of a multi-loop architecture. For example a large β_1 value saturates the DCO bank control word reducing the effectiveness of a high $\beta_1 K_{T1}$ gain. The case is illustrated by the blue curve in Fig. 9 where the coarse bank has just 32 capacitive cells. To overcome this limitation, the adoption of a feedforward path, highlighted in blue in Fig. 10, is proposed.

A. Feed-Forward Locking Transient

Fig. 10 shows the corresponding model for transient analysis. The divider FCW can handle integer and fractional N values with a small residual quantization. By controlling the fractional part of FCW, a fractional part of the T_{DCO} period can be added or subtracted to the signal t_d , thus acting like

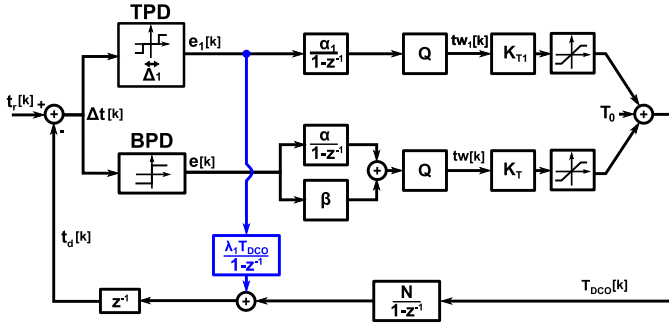


Fig. 10. Model of the BB-DPLL with a frequency-aid feed-forward branch.

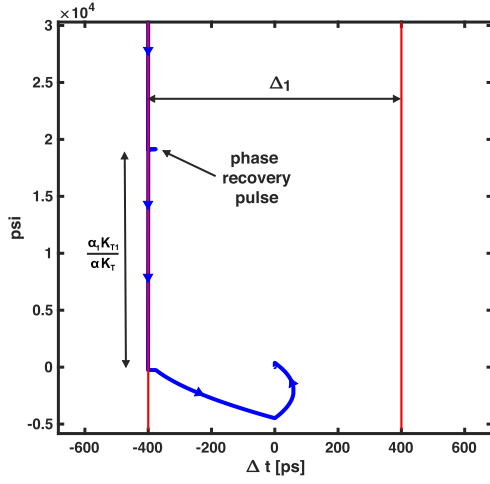


Fig. 11. State-space representation of a locking trajectory showing the effect of the phase recovery pulse.

the intrinsic integrator of the DCO path. Therefore, the feed-forward branch is equivalent to the proportional branch in the multi-loop system, but it has a much lower quantization and no link with the DCO sensitivity. Based on the equivalence of the two schemes we can write

$$T_0 \lambda_1 = \beta_1 K_{T1} N. \quad (12)$$

Therefore, to synthesize the same β_1 of Section II, the λ_1 value should be

$$\lambda_1 = \frac{K_{T1} N \beta_1}{T_0}, \quad (13)$$

leading to $\lambda_1 \approx 2^{-9}$. The resulting trajectory is the one already described in Fig. 6, showing the same locking behaviour of Fig. 8. In this way, the proportional feed-forward branch can implement high β_1 values without incurring into saturation effects. Fig. 9 shows the comparison with the multi-loop approach by increasing λ_1 and therefore the equivalent β_1 value, given by

$$\beta_{1,eq} = \frac{\lambda_1 T_0}{K_{T1} N}. \quad (14)$$

In addition, the analysis highlights a new type of locking trajectory, which is much faster in settling than the one analyzed so far. In Appendix C it is shown that, for frequency

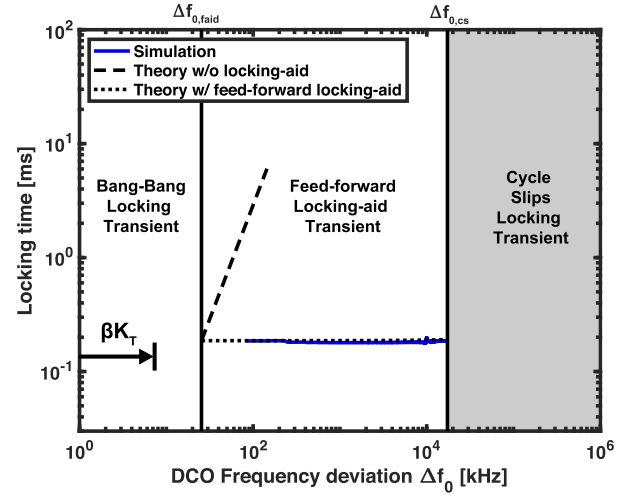


Fig. 12. Comparison between the locking time of a BB-DPLL with multi-loop locking-aid (dashed line) and with feed-forward locking-aid architecture (dotted line). The blue line shows the simulation results.

jumps Δf less than

$$\Delta f_{th} \approx \lambda_1 \frac{f_0}{N} = \lambda_1 f_{ref}, \quad (15)$$

the transient follows the features in Fig. 11. When the locking-aid is activated, the trajectory is pushed back inside the dead-zone, reducing Δt with a phase recovery pulse, while the integration variables, Ψ and Φ , and therefore ΔT_0 , are decreased at each reference cycle. The recovery trajectory is trapped close to the dead-zone boundaries until the external locking-aid path is not triggered anymore. This behaviour, where the phase error is instantaneously reduced, by a phase recovery pulse, can be denoted as *phase recovery trajectory* in contrast to the *frequency recovery trajectory* presented in the previous section. In Appendix C, starting from the consideration that ΔT_0 is reduced by $\alpha_1 K_{T1}$ at each TPD activation, it is shown that the locking time in phase recovery mode can be approximated as:

$$T_{lock} \approx T_{ref} \frac{1}{R(2-R)} \left(\frac{\Delta T_{0,faid}}{\beta K_T} \right)^2 + 2T_{ref} \frac{(\Delta T_0 - \Delta T_{0,faid})}{\alpha_1 K_{T1}}. \quad (16)$$

Note that the second term, due to the locking aid loop, instead of being quadratic on the initial error as in (11), becomes proportional to the difference between the DCO frequency deviation ΔT_0 and the threshold $\Delta T_{0,faid}$.

A similar approach of phase compensation was presented in [21]. But in this case the feed-forward branch directly controls the divider module in an integer charge-pump PLL. In a digital non-linear discrete-time system, the locking transient cannot be studied with linear approximation as proposed in [21], but the locking trajectory should be taken into account.

Fig. 12 shows the locking time of the system with the feed-forward branch compared to the single loop BB-DPLL with no frequency aid. The value of λ_1 is around 2^{-3} , that is equivalent, using (14), to $\beta_{1,eq} \approx 2^5$. The dead-zone is set to $\Delta_1 = 800$ ps and $R_1 = 2^{-6}$ also for this

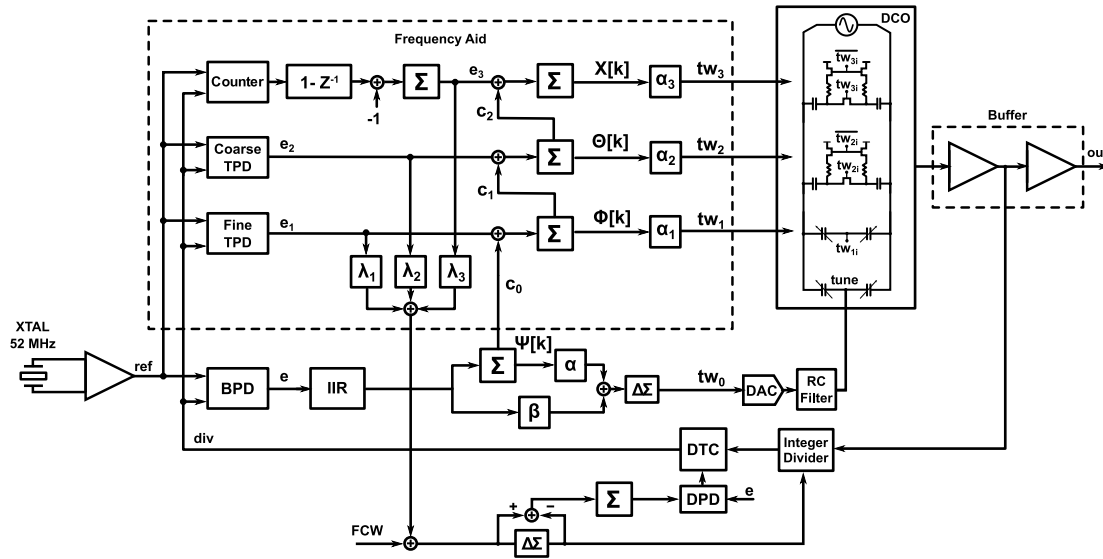


Fig. 13. Scheme of the complete system: main BB-DPLL with all the frequency aid branches.

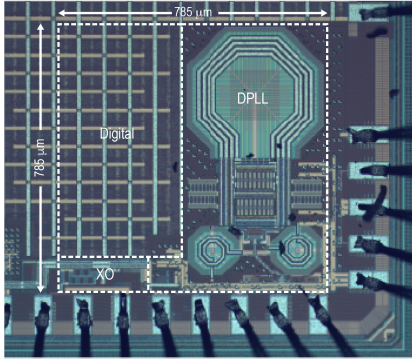


Fig. 14. Die Microphotograph.

topology. The frequency deviation enabling the locking-aid path is $\Delta f_{0, \text{faid}} \approx 26 \text{ kHz}$. The quantization of coarse capacitive bank is considered. The simulated results shown by the blue curve are slightly below the estimate derived from (16). The phase recovery regime holds up to a frequency error threshold $\Delta f_{0, \text{th}} \approx 5 \text{ MHz}$. Beyond this value, the usual frequency recover regime takes place and cycle slips occur for $\Delta f > \Delta f_{0, \text{cs}} \approx 17.3 \text{ MHz}$.

Note that, even if the cycle slip window has been greatly extended with respect to the initial value of 2.2 MHz , $\Delta f_{0, \text{cs}} \approx 17.3 \text{ MHz}$ is only 0.45% of the DCO central frequency. It follows that, using DCOs with large tuning range, frequency steps within the DCO tuning range may start a cycle slip. To mitigate this limitation, a new locking-aid loop can be added with a coarse TPD having a larger dead-zone Δ_2 and a state variable $\Theta[k]$. Fig. 13 represents the whole system. The additional locking-aid path provides more degrees of freedom and a larger $\alpha_2 K_{T2}$ to further enlarge the cycle slip threshold beyond the DCO tuning range. A possible implementation of the two TPDs with dead-zones Δ_1 and Δ_2 is shown in Fig. 15. On one branch the *div* signal is delayed and compared

with the *ref* signal, on the other branch the *ref* signal is delayed and compared with the *div* signal. The output of the four comparisons, performed with lead-lag flip-flops are then encoded by a look-up table to provide the TPD characteristic also presented in Fig. 15. The power consumption of this structure is lower than the power consumption of a standard counter or of a finite state machine solution being of the order of $10 \mu\text{W}$, proportional to the capacitance loading each delay stage. However, despite all these refinements, cycle slips may take place due to the saturation of the control word of the DCO bank. The locking-aid architecture with nested dead zones must be therefore extended by adding a mechanism able to recover these specific situations. In a cycle slip the DCO frequency is controlled in the wrong direction, thus generating a large frequency difference between reference and divider signals. To detect this condition a simple counter, clocked by the falling edge of the reference signal, is resampled by the rising edge of the divider signal. The difference between the current output of the resampled counter and the previous value gives how many reference periods fall within a divider period. In lock condition only one reference falling edge is present. If the frequencies are instead different, each cycle accumulates a phase shift until two or zero reference falling edges are detected into a divider cycle. It follows that, if the output of the derivative block following the counter is 0 the divider frequency is higher than the reference one. If instead the output is 2, the divider frequency is lower.

To properly control the ultra-coarse tuning bank of the DCO an additional path is therefore added after the counter, as depicted in the complete PLL scheme in Fig. 13. The counter compares the reference and the divider phases. The derivative block extracts the number of reference edges within a divider period. The result, shifted by one, gives the useful information. Summing up these values with an integrator the cumulative phase error $e_3[k]$ is recovered and is used to control the DCO output as in the other TDC paths. To reduce

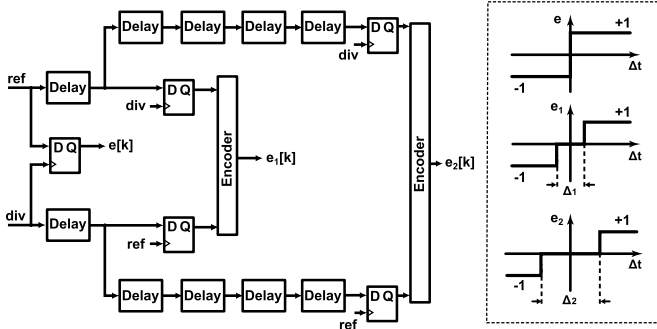


Fig. 15. Schematic of BB-PD, Fine TPD and Coarse TPD.

the power consumption, a gray coded counter was adopted, optimized for reducing the switching activity during the steady state of the PLL.

In summary, the frequency aid block is composed by the two TPDs depicted in Fig. 15 and a counter. These three blocks are used to generate the three lead-lag error signals, $e_1[k]$, $e_2[k]$ and $e_3[k]$. The greater the phase error the greater the number of error signals triggered. This information is integrated by accumulators to convert it into a frequency control information fed to the DCO capacitive banks. The three accumulators, with $\Phi[k]$, $\Theta[k]$ and $X[k]$ as outputs, are linked together and to the main loop integral path accumulator with output $\Psi[k]$ by the carry signals $c_0[k]$, $c_1[k]$, $c_2[k]$. For example, once the output $\Phi[k]$ saturates the DCO capacitive bank controlled by $tw_1[k]$, the carry signal $c_1[k]$ increases the control of the larger capacitive bank $tw_2[k]$ while the value of $\Phi[k]$ resets. The procedure holds also for the other accumulators and guarantees the tuning range continuity of the DCO despite bank segmentation. The parameters α_1 , α_2 , α_3 and λ_1 , λ_2 , λ_3 are used to control the speed of the three frequency aid branches based on the equations derived in Sections III and IV.

V. EXPERIMENTAL RESULTS

The frequency synthesizer was fabricated in 65-nm CMOS process [15]. Fig. 14 shows the die microphotograph of the implemented system. An integrated Pierce oscillator is connected to an external FA-118T EPSON quartz crystal generating the 52-MHz reference signal. The complementary Class-B oscillator with resonant tails at both transconductors common node can be controlled from 3.7 GHz to 4.1 GHz, corresponding to a tuning range of 12%. The tuning range was subdivided into three switched capacitive banks plus an analog varactor. The period gains and the number of levels for each bank are: $K_T = 1.11 fs/LSB$ and 32 levels for the analog varactor, $K_{T1} = 10.9 fs/LSB$ and 32 levels for the fine bank, $K_{T2} = 175.3 fs/LSB$ and 32 levels for the coarse bank, $K_{T3} = 2805 fs/LSB$ and 16 levels for the ultra-coarse bank.

The flicker noise corner frequency is 60 kHz, close to the simulated one. A double stage Current Mode Logic (CML) buffer isolates the DCO from the output PAD driver and from the divider block, relaxing the load directly connected to the LC tank. The total power dissipation is 5.28 mW leading to a

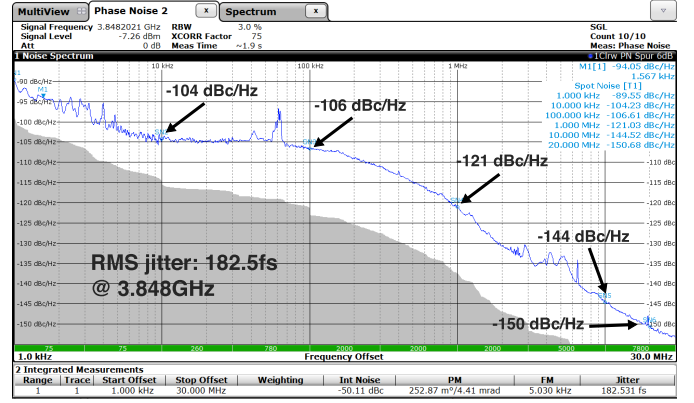


Fig. 16. Phase Noise of the 3.8 GHz output signal.

FoM of -247.5 dB. Fig. 16 shows the measured phase noise. The *RMS* jitter (integrated from 1 kHz to 30 MHz) is 182.5 fs, while the spot phase noise at 20 MHz offset is -150.7 dBc/Hz. This value meets the tight GSM specifications referred to a 900MHz carrier of -162 dBc/Hz.

These results are obtained thanks to the high efficiency and low noise oscillator design, similar to the one proposed by [18], combined with a low noise reference chain. The quantization noise coming from the BB-PD is reduced by a digital IIR filter with a pole at 2.6 MHz, as in [17], while the quantization noise at the input of the DCO, due to the delta sigma modulator that controls the DAC, is filtered-out by a second order RC low-pass analog filter with poles at 2.5 MHz and 10 MHz. Since both the IIR and the RC filter poles are outside the PLL bandwidth (150kHz) they do not affect the locking transient of the main loop.

The FCW is generated by a sigma delta modulator that drives the integer divider and the DTC, through a Digital Pre-Distortion (DPD) block. The DPD estimates the correlation between the digital phase error $e[k]$ and the quantization error of the sigma delta. This estimation is used to calibrate the word fed to the DTC to obtain a complete cancellation of the divider sigma-delta quantization error. Once the DTC is calibrated the PLL can reliably operate in fractional synthesis mode.

Phase noise and locking transient measurement were performed using R&S FSWP-50 and Anritsu MG2690A signal and phase noise analyzer. Transient performance was tested for both small (fractions of the reference frequency) and large (integer multiples of the reference frequency) frequency steps to prove the validity of the implemented algorithm. The locking transient was induced by swiftly changing the FCW while the output frequency was measured to detect when the PLL reaches the new steady state condition. The blue line Fig. 17 shows the locking time measured by using only the steady state loop designed for the optimum jitter obtained in Fig. 16. Without any locking-aid techniques this loop is unable to lock for a frequency steps larger than 200 kHz. The dashed line represents the theoretical locking time in the same condition computed using (3). The red curve shows instead the locking time reached by using the locking-aid techniques. A frequency

TABLE I
COMPARISON TABLE WITH OTHER FAST-SETTLING DPLLs

	This work	[23]	[17]	[24]	[25]	[26]	[27]
Architecture	BB-DPLL	DPLL	BB-DPLL	DPLL	DPLL	BB-DPLL	BB-DPLL
CMOS process	65 nm	130 nm	65 nm	40 nm	65 nm	28 nm	28 nm
Ref. Freq. [MHz]	52	65	40	32	50	216	500
Out. Freq. [GHz]	3.7-4.1	1.56	2.9-4.0	2.1-2.7	1.0-2.0	22.5 -27.7	12.8-15.2
Tuning Range [%]	12	0.58	31.8	25	66	20.7	17
RMS jitter [fs]	183	385	400	1710	3090	220	66.2
Power [mW]	5.28	2.8	4.5	0.86	10.8	25	19.8
FoM [†] [dB]	-247.5	-244	-241.4	-236	-219.9	-239	-250.6
BW [MHz]	0.15	0.07	0.7	2	n.a.	0.2	2
Freq. Hop [MHz]	364	1	40	1.5	<1000	864	1000
T_{lock} [μ s]	115	150	120	16	<0.7	115	18.55
Settling accuracy [ppm]	90	n.a.	n.a.	n.a.	n.a.	3	70
T_{lock} [ref cycle]	5980	9750	4800	512	3090	24840	9275
Area [mm^2]	0.61	1.06	0.22	0.2	0.044	0.09	0.17

[†] FoM=10log($\sigma/1s$)²($P_{mW}/1mW$)

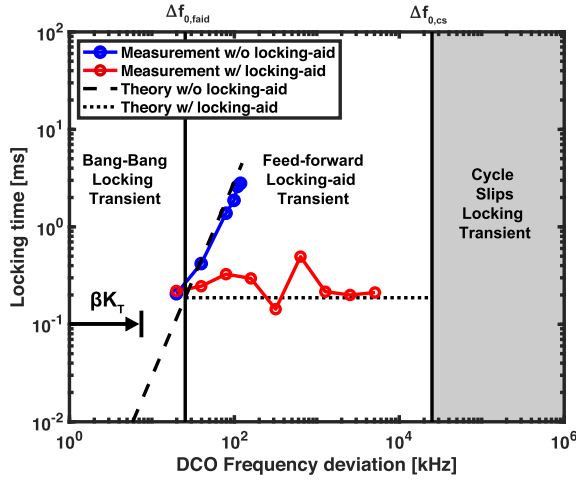


Fig. 17. Measured BB-DPLL locking transients: without locking-aid (blue line) and with the locking-aid technique (red line) for frequency steps smaller than 10 MHz. The locking time values are taken for a settling within 10 kHz from the target frequency.

step around 26 kHz is able to switch the locking-aid branch. The related theoretical locking time using (16), $\alpha_1 = 2^{-1}$ and $\lambda_1 = 2^{-3}$ is highlighted by the dotted line, running close to the measured values. Fig. 18 shows the transient response to a coarse frequency acquisition measured by (a) disabling and (b) enabling the frequency aid technique, respectively. In this case the tested prototype had a smaller dead-zone with respect to the one characterized in Fig. 17, proving that the feed-forward topology is effective also with different dead-zone and wide frequency steps. The counter block was always enabled to guarantee the frequency locking. In the first row a fine frequency step of 1 MHz is performed. Without the frequency aid branch the transient is heavily nonlinear and the target frequency is reached after 7 ms. Instead, with the two frequency aid branches active, the lock condition is achieved in just 110 μ s when the frequency settles within 10 kHz from the target value. In the second row of Fig. 18 a step of 364 MHz is performed. Disabling the frequency-aid technique (while keeping the frequency discriminator block), the circuit

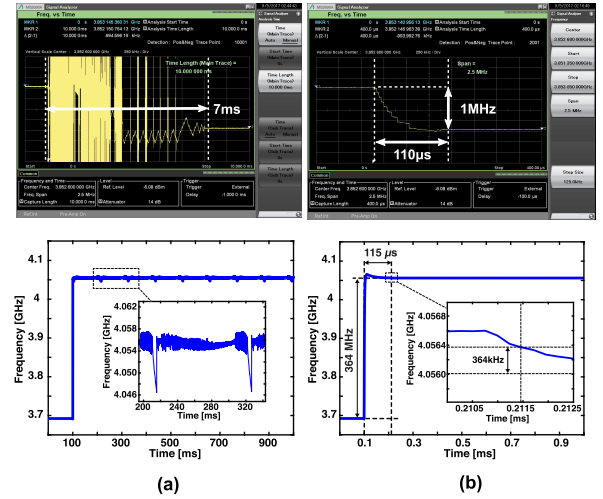


Fig. 18. Measured BB-DPLL locking transients: (a) without frequency aid and (b) with the frequency aid technique for 1 MHz (top) and 364 MHz frequency steps (bottom).

is unable to lock due to cycle slips. Enabling the frequency aid technique, the BB-DPLL locks in 5.6 μ s within 10 MHz from the target frequency value and takes 115 μ s to fall below 364 kHz, equivalent to an accuracy of 90 ppm of the target frequency.

Table I shows a comparison with other fast-settling DPLLs. Note that as the PLL bandwidth decreases to improve jitter performance, the locking time for coarse frequency hops become a key design challenge. The multi-loop architecture presented in the paper is effective in retaining very low-jitter at steady state, low power consumption (very good FoM) while providing fast-settling within a tight accuracy interval both as absolute value and as number of reference cycles.

VI. CONCLUSION

The BB-DPLL architecture is increasingly used in the field of high performance frequency synthesizers due to its scalability and low power consumption with respect to the standard multi-bit phase detector PLLs. Unfortunately, the

discrete nature and the limited dynamic range of the BB-PD output impair the locking time. The paper presents a new locking-aid scheme that exploiting a multi-loop DCO control and a feed-forward path to improve the locking time without degrading the noise performance of the system. With this approach and the proper sizing of dead-zone and locking-aid loop parameters is possible to avoid cycle slips also for large frequency steps while retaining low jitter at steady state.

APPENDIX A

RE-LOCKING TIME IN A SINGLE LOOP BB-DPLL

Referring to Fig. 3, let us assume that for $k < 0$, the PLL is at steady state in a random noise regime [19], [20]. Being a type-II PLL, the average values of $\Delta t[k]$ and $e[k]$ are zero. At $k = 0$ the modulus of the divider changes from N to $N + 1$ while the reference frequency f_{ref} remains constant ($T_r[k] = T_{ref} = 1/f_{ref}$). The loop begins to adjust the DCO period, T_{DCO} to reach again a lock condition. The initial value is $T_{DCO}[-1] = T_{ref}/N$, while in the final steady state condition $T_{DCO}[+\infty] = T_{ref}/(N + 1)$.

To simplify the discussion let us assume that $T_{DCO}[+\infty] = T_0$. It follows that $T_{DCO}[-1] = T_0 + T_0/N$ and $T_{ref} = NT_{DCO}[-1] = (N + 1)T_{DCO}[+\infty] = (N + 1)T_0$. The assumption has no impact on the generality of the results that will only depend on the distance between the initial and final loop states and by the loop parameters. For $k = 0$ the state variable $\Psi[0]$ is derived from the frequency lock condition

$$N(\Psi[0]\alpha K_T + T_0) = T_{ref} = (N + 1)T_0, \quad (17)$$

leading to

$$\Psi[0] = \frac{T_0}{N\alpha K_T}, \quad (18)$$

At the end of the locking transient, the state variable $\Psi[k]$ reaches the value $\Psi[\infty] = 0$. $\Psi[\infty]$ can be linked to the loop parameters by writing the frequency lock condition at $k = +\infty$. It is

$$(N + 1)(\Psi[\infty]\alpha K_T + T_0) = T_{ref}. \quad (19)$$

Since $\Psi[\infty] = 0$, it turns out $T_{ref} = (N + 1)T_0$, as expected. Regarding the average time error, being the loop a type-II PLL, $\overline{\Delta t} = 0$ also at the end of the transient when the new steady state is reached.

An alternative approach is to consider the divider modulus fixed to N and the reference frequency changing from an initial period $T_r[-1] = (N + 1)T_0$ to $T_r[0] = T_r[+\infty] = NT_0$. It is easy to realize that the state variables $\Psi[k]$ and $\Delta t[k]$ for $k = -1$ and $k = +\infty$ remain the same. Therefore, the transient follows the same locking trajectory. The only difference being that the latter case is more convenient to analyze since the PLL model has constant loop parameters.

Due to the change of the reference period, the DCO output period $T_{DCO}[-1] = (T_0 + T_0/N)$ is longer than its final steady state value, T_0 , thus leading to a negative phase error $\Delta t[0] < 0$. The BB-PD digital output becomes $e[0] = -1$ and the state variable $\Psi[k]$ is decreased. To quantitatively derive the $\Delta t[k]$ transient, note that at $k = 0$, both $t_r[0]$ and

$t_d[0]$ are zero. After the change of the reference period from $(N + 1)T_0$ to NT_0 , it is

$$\begin{aligned} t_r[0] &= 0 \\ t_r[1] &= NT_0 \\ t_r[2] &= 2NT_0 \\ t_r[k] &= kNT_0, \end{aligned} \quad (20)$$

while

$$\begin{aligned} t_d[0] &= 0 \\ t_d[1] &= \alpha\Psi[0]NK_T + NT_0 \\ t_d[2] &= [\alpha(\Psi[0] - 1) - \beta]NK_T + NT_0 + t_d[1] \\ t_d[k] &= \beta K_T N + \sum_{i=0}^{k-1} [\alpha(\Psi[0] - i) - \beta]NK_T + kNT_0 \end{aligned} \quad (21)$$

These equations consider a noiseless PLL but in reality the noise affects the zero crossing. Noise adds an additional $\pm\beta K_T N$ at the timestamp $t_d[1]$ and the generic equation $t_d[k]$ should also include this term. However, for practical analysis, since the $\beta K_T N$ term is small this contribution can be neglected. It follows that during the first section of the transient $\Delta t[k] = t_r[k] - t_d[k]$ is negative and given by

$$\begin{aligned} \Delta t[k] &= -\beta K_T N - \sum_{i=0}^{k-1} [\alpha(\Psi[0] - i) - \beta]NK_T \\ &= \left[(-\alpha\Psi[0] + \beta)k - \frac{k\alpha}{2} + \frac{k^2\alpha}{2} \right] K_T N - \beta K_T N. \end{aligned} \quad (22)$$

To use a more compact notation we can consider for the final steady state $\Psi_\infty = \Psi[\infty] = 0$ while the value $\Psi_0 = \Psi[0]$ corresponds to the distance between the initial state and the origin of the state-space representation. In the same way Ψ_m will represent the state variable value Ψ at the beginning of the m -th section. Using these notations and neglecting the last term, Δt from (22) can be written as

$$\Delta t[k] \simeq \left[\left(-\Psi_0 + \frac{1}{R} - \frac{1}{2} \right) k + \frac{k^2}{2} \right] \alpha K_T N. \quad (23)$$

The number of steps required to reach a zero phase error can be derived from (23) as

$$k_1 = 2\Psi_0 - \frac{2}{R} + 1. \quad (24)$$

At $k = k_1 + 1$ the phase error $\Delta t[k_1 + 1]$ becomes positive and the BB-PD output switches to $e[k] = +1$. The starting point Ψ_1 for the new branch of the trajectory is

$$\Psi_1 = \Psi_0 - \left(2\Psi_0 - \frac{2}{R} + 1 \right) = -\Psi_0 + \frac{2}{R} - 1. \quad (25)$$

To simplify, since $\Delta t[k_1] = 0$, the time axis can be shifted, thus starting again from $k = 0$. The trajectory equation is now changed, being $e[k] = +1$, thus leading to

$$\Delta t[k] = \beta K_T N - \sum_{i=0}^{k-1} (\alpha(\Psi_1 + i) + \beta) K_T N, \quad (26)$$

$$\Delta t[k] \simeq \left[\left(-\Psi_1 - \frac{1}{R} + \frac{1}{2} \right) k - \frac{k^2}{2} \right] \alpha K_T N, \quad (27)$$

which becomes nil again after k_2 steps, where

$$k_2 = -2\Psi_1 - \frac{2}{R} + 1 = 2\Psi_0 - 3\left[\frac{2}{R} - 1\right]. \quad (28)$$

The transient goes on until the final steady state is reached (Fig. 3).

Note that at the end of the m -th trajectory section, the distance between the current state, Ψ_m , and the final state Ψ_∞ is always reduced by

$$\Delta\Psi = (2/R - 1). \quad (29)$$

Therefore the number of trajectory sections needed to reach the transient end is $\Psi_0/\Delta\Psi$. Taking $m = 0$ for the first transient section, based on (24) and (26), the number of reference cycles to complete the m -th trajectory section can be written as $2\Psi_0 - (2m + 1)\Delta\Psi$, and the total amount of steps to complete the transient is derived as

$$total_steps = \sum_{m=0}^{\frac{\Psi_0}{\Delta\Psi}-1} 2\Psi_0 - (2m + 1)\Delta\Psi \approx \frac{\Psi_0^2}{\Delta\Psi}. \quad (30)$$

Moreover, the value Ψ_0 giving to the distance of the initial state from the origin of the state-space is linked to the difference ΔT_0 between the initial DCO period and the steady state T_0 value

$$\Delta T_0 = \alpha K_T \Psi_0. \quad (31)$$

Substituting (29) and (31) in (30) and multiplying the result by the duration of a reference clock cycle, T_{ref} , we obtain the locking time

$$T_{lock} \approx T_{ref} \frac{1}{R(2-R)} \left(\frac{\Delta T_0}{\beta K_T} \right)^2, \quad (32)$$

which is the equation in (3).

Moreover, by taking the derivative of (27), it can be found that the peak time error is reached for

$$k_{max} = -\Psi_1 - \frac{1}{R} + \frac{1}{2}, \quad (33)$$

and its value is

$$\Delta t_{max} = \frac{1}{2} \left(-\Psi_1 - \frac{1}{R} + \frac{1}{2} \right)^2 \alpha K_T N \approx \frac{\Psi_1^2}{2} \alpha K_T N, \quad (34)$$

under the assumption of a large DCO frequency deviation. More in general, denoting as $\Delta T_m = \alpha K_T \Psi_m$ the distance of the DCO period from the steady state value at the beginning of the m -th trajectory section, we get

$$\Delta t_{max} = \frac{\alpha K_T N}{2} \left(\frac{\Delta T_m}{\alpha K_T} \right)^2, \quad (35)$$

which is used to write (2).

APPENDIX B LOCKING TIME CALCULATION WITH A FREQUENCY-AID LOOP

Let us denote as $\Phi[k]$ the state variable of the of the integral path of the locking-aid branch. Starting from the trajectory

equation (27) written for the first locking section $m = 0$, and solving it for $\Delta t = \Delta_1/2$, we obtain the second order equation

$$-\frac{k_{faid}^2}{2} + \left(-\Psi_0 - \frac{1}{R} + \frac{1}{2} \right) k_{faid} - \frac{\Delta_1}{2\alpha K_T N} = 0. \quad (36)$$

Its first root, which is the instant k_{faid} , is given by

$$k_{faid} \approx \Psi_0 - \sqrt{\Psi_0^2 - \frac{\Delta_1}{\alpha K_T N}}. \quad (37)$$

The state variable Ψ_1 , is given by the initial DCO state Ψ_0 minus the number of steps of the trajectory section, that is

$$\Psi_1 = \Psi_0 - k_{faid} \approx \sqrt{\Psi_0^2 - \frac{\Delta_1}{\alpha K_T N}}, \quad (38)$$

which can be also written as

$$\Psi_1 \approx \sqrt{\left(\frac{\Delta T_0}{\alpha K_T} \right)^2 - \left(\frac{\Delta T_{0,faid}}{\alpha K_T} \right)^2}, \quad (39)$$

where ΔT_0 is the error of the DCO period at the beginning of the transient and $\Delta T_{0,faid}$ is given by (10).

The calculated state variable Ψ_1 is the starting point of the locking section driven by the frequency-aid loop. When the frequency aid loop comes into play, the DCO time error is given by

$$\Delta T_{DCO}[k] = (\Psi[k]\alpha K_T + \Phi[k]\alpha_1 K_{T1}) = \alpha_1 K_{T1} M_1[k], \quad (40)$$

where, to have a compact notation, we define

$$M_1[k] = \left(\Psi[k] \frac{\alpha K_T}{\alpha_1 K_{T1}} + \Phi[k] \right). \quad (41)$$

At $k = k_{faid}$ the trajectory starts from $\Delta t[k_{faid},0] = \Delta_1/2$. After shifting the y-axis to the dead-zone edge ($\Delta t_{faid}[k] = \Delta t[k] - \Delta_1/2$) and making, the k index starting from zero, the trajectory equation becomes the same already described in (22)

$$\Delta t_{faid}[k] \approx - \sum_{i=0}^{k-1} \left[M_1[0] - \frac{\beta_1}{\alpha_1} - i \right] K_{T1} \alpha_1 N, \quad (42)$$

where $\Phi[k_{faid}] = 0$ and $M_1[0] = \Psi[k_{faid}](\alpha K_T / \alpha_1 K_{T1})$.

With an initial condition $M_1[0]$ the number of steps k_1 to reach $\Delta t_{faid} = 0$ is similar to (24)

$$k_1 \approx 2M_1[0] - \left(\frac{2}{R_1} - 1 \right). \quad (43)$$

As Δt_{faid} becomes nil again, the trajectory enters the dead-zone region and the internal loop takes over until Δt overcomes again the dead-zone boundary. The crossover between the dead-zone edges may happen many times until the residual DCO period error becomes less than the critical value $\Delta T_{0,faid}$. At this point the frequency aid path will not be further triggered and the PLL reaches the random noise regime in T_{lock0} , as defined in (32)

$$T_{lock0} \simeq T_{ref} \frac{1}{R(2-R)} \left(\frac{\Delta T_{0,faid}}{\beta K_T} \right)^2. \quad (44)$$

To give a manageable expression of the total locking time, let us neglect the first part the transient (37) and the transitions crossing the dead-zone. Under these simplifying assumptions, the total locking time is written as the sum of the critical locking time T_{lock0} from (44) and the time spent in a frequency aid trajectory, T_{lock1} , using (30) with the locking-aid parameters. It is

$$T_{lock1} \approx T_{ref} \cdot tot_steps_f_{aid} \approx T_{ref} \frac{M_1[0]^2}{\Delta M_1}, \quad (45)$$

with $\Delta M_1 = 2/R_1 - 1$. That is

$$t_{lock} \approx T_{ref} \frac{1}{R(2-R)} \left(\frac{\Delta T_{0,f_{aid}}}{\beta K_T} \right)^2 + T_{ref} \frac{1}{R_1(2-R_1)} \frac{\Delta T_0^2 - \Delta T_{0,f_{aid}}^2}{(\beta_1 K_{T1})^2}. \quad (46)$$

APPENDIX C

FEED-FORWARD PHASE RECOVERY LOCKING TRANSIENT

Let us derive here an estimate of the locking time limited by the phase recovery transient shown in Fig. 11. To this aim refer to the model in Fig. 10. Taking $N - \lambda_1 \approx N$, the first step just after the frequency aid activation has a Δt that can be written as

$$\Delta t [1] = -(M_1 - 1)K_{T1}\alpha_1 N + \lambda_1 T_0. \quad (47)$$

If the $\Delta t [1]$ value is negative, the trajectory follows the parabolic path already described before in Appendix B. If instead the $\Delta t [1]$ value is positive, the frequency-aid path is immediately disabled, while the phase recovery pulse through the feed-forward path helps to reduce the phase error pushing the PLL under the control of the inner loop.

The boundary ΔT_{th} between negative and positive $\Delta t[1]$ values is given by the condition

$$(M_1 - 1)K_{T1}\alpha_1 N = \lambda_1 T_0. \quad (48)$$

Replacing Ψ_1 given by (39) into (41), we obtain the M_1 value and (48) becomes

$$N\sqrt{\Delta T_{th}^2 - \Delta T_{0,f_{aid}}^2} = \lambda_1 T_0. \quad (49)$$

For large period deviations ($\Delta T_{th} \gg \Delta T_{0,f_{aid}}$) and using $\Delta f/f_0 = \Delta T/T$, we get that the phase recovery transient is triggered provided that the frequency step is less than

$$\Delta f_{th} \approx \lambda_1 \frac{f_0}{N} = \lambda_1 f_{ref}. \quad (50)$$

Starting from a ΔT_0 period error, the number of locking-aid activations needed to reach the edge of inner dead-zone is

$$N_{f_{aid}} = \frac{\Delta T_0 - \Delta T_{0,f_{aid}}}{\alpha_1 K_{T1}}, \quad (51)$$

and the locking time can be therefore approximated as:

$$T_{lock} \approx T_{ref} \frac{1}{R(2-R)} \left(\frac{\Delta T_{0,f_{aid}}}{\beta K_T} \right)^2 + 2T_{ref} \frac{(\Delta T_0 - \Delta T_{0,f_{aid}})}{\alpha_1 K_{T1}}. \quad (52)$$

where the factor 2 multiplying the last term turns out taking into account that in this regime each activation of the frequency aid branch is followed by a cycle falling within the dead zone (see Fig. 11).

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